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رياضيات متقطعة ٧
هندسة، انذكار، استاذ

الرياضيات المتقطعة العددية

المحاضرة السابعة

المرحلة الثالثة

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1. Introduction

Graphs are fundamental mathematical structures used to model pairwise relationships between objects. They form the backbone of many artificial intelligence applications, including recommendation systems, pathfinding, network analysis, and social network modeling.

A graph consists of:

Vertices (Nodes) •

Edges which may be: •

Directed or Undirected ◦

Weighted or Unweighted ◦

2. Categories of Graph Algorithms

Graph algorithms fall into three main categories:

A. Graph Traversal Algorithms

1. Breadth-First Search (BFS)

Uses a Queue •

Explores nodes level-by-level •

Works best for **unweighted graphs** •

Applications:

Finding the **shortest path** in unweighted graphs •

• Social network analysis •

• Recommendation systems •

2. Depth-First Search (DFS)

• Uses a Stack •

• Explores one full path before backtracking •

• Used in: •

◦ Cycle detection ◦

◦ Maze solving ◦

◦ Connected components ◦

◦ Topological sorting ◦

B. Shortest Path Algorithms

1. Dijkstra's Algorithm

Finds the shortest path in positively weighted graphs •

Uses a priority queue •

Applications:

Google Maps navigation •

Network routing •

Robot path planning •

2. Bellman-Ford Algorithm

Handles negative weights •

Detects negative cycles •

Slower than Dijkstra's algorithm •

3. Floyd-Warshall Algorithm

Finds shortest paths between all pairs of nodes •

Uses dynamic programming •

Complexity: $O(n^3)$ •

Applications:

Large network systems •

Transportation and logistics planning •

Cost analysis •

C. Minimum Spanning Tree (MST) Algorithms

1. Kruskal's Algorithm

- Sorts edges by weight
- Builds MST using Union-Find data structure

2. Prim's Algorithm

- Starts from any node
- Repeatedly selects the minimum-weight edge connecting the tree to a new node

Applications of MST:

- Designing electrical networks
- Reducing road construction cost
- Internet network design
- Clustering in AI

3. Comparison Table

Best Algorithm	Notes
BFS	Simple and fast
Dijkstra	Very efficient
Bellman-Ford	Handles negative weights
Floyd-Warshall	Dynamic programming
Prim / Kruskal	Builds MST

4. Solved Examples

Example 1: BFS Traversal

Graph:

A — B — C

| |

D — E — F

Starting from A:

Order of visitation (BFS):

A, B, D, C, E, F

Example 2: DFS Traversal

Possible DFS order:

A → B → C → F → E → D

(The order may vary depending on neighbor ordering.)

Example 3: Dijkstra's Algorithm

Example

Graph weights:

$A \rightarrow B$ (2)

$B \rightarrow C$ (3)

$A \rightarrow C$ (5)

$B \rightarrow D$ (1)

$C \rightarrow D$ (2)

Find shortest path from A to D:

Start: $A = 0$ •

Neighbors: $B=2, C=5$ •

Choose B (2) $\rightarrow$ Update $D = 3$ •

Choose D (3) $\rightarrow$ Done •

Shortest Path: $A \rightarrow B \rightarrow D$

Cost = 3

5. Applications in Artificial Intelligence

Graph algorithms are widely used in AI:

1. Robotics

- Path planning •

- Obstacle avoidance •

2. Social Network Analysis

- Community detection •

- Influence measurement •

- Friend recommendation •

3. Search Engines

- PageRank algorithm is graph-based •

4. Recommendation Systems

- Identifying relationships between users and items •

5. Game AI

- Map navigation •

- Decision tree exploration •

6. Questions for Students

1. Draw a weighted graph of 5 cities and find the shortest path between two selected cities.

2. Explain the difference between BFS and DFS and give a real-world example of each.

3. Why is MST different from shortest path algorithms?

السؤال الثاني بطريقة BFS

السؤال الأول بطريقة MST